Hunmin (Troy) Lee Ph.D. Candidate

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Selected Publications

1. **Hunmin Lee**, B.L, M.J., Z.Y., Q.Z. Few-shot prototype adaptation for generalizable electromyography gesture recognition. *Scientific reports (IF: 3.9)*, 2026.
2. **Hunmin Lee**, K.W., D.S. FedPML: Federated Prototype Meta-Learning for Adaptation and Generalization in Distributed Environments. *IEEE Internet of Things Journal (IF: 9.9)*, 2026.
3. **Hunmin Lee**, M.J., Q.Z. FedEMG: Achieving Generalization, Personalization, and Resource Efficiency in EMG-based Upper-Limb Rehabilitation through Federated Prototype Learning. *IEEE Transactions on Biomedical Engineering (IF: 4.5)*, 2025.
4. **Hunmin Lee**, D.S. Federated Prototype Adaptation Learning: Personalized and Generalizable Learning in Distributed IoT Networks. *IEEE Internet of Things Journal (IF: 9.9)*, 2025.
5. **Hunmin Lee**, H.S., J.B., W.K., H.K., D.S. FedVaccine: Robust Federated Learning in Noisy and Non-IID Wireless Network Environments. *IEEE Transactions on Network Science and Engineering (IF: 7.9)*, 2025.
6. **Hunmin Lee**, M.J., J.Y., Z.Y., Q.Z. Decoding gestures in electromyography: spatiotemporal Graph neural networks for generalizable and interpretable classification. *IEEE Transactions on Neural Systems and Rehabilitation Engineering (IF: 5.7)*, 2024.
7. **Hunmin Lee**, M.J., Q.Z. EMGCipher: Decoding electromyography for upper-limb gesture classification with explainable AI for resource optimization. *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024. (**Oral**)
8. **Hunmin Lee**, M.J., Q.Z. FedAssist: Federated Learning in AI-Powered Prosthetics for Sustainable and Collaborative Learning. *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024.
9. **Hunmin Lee**, M.J., Q.Z. Unveiling EMG semantics: a prototype-learning approach to generalizable gesture classification. *Journal of Neural Engineering (IF: 3.8)*, 2024.
10. **Hunmin Lee**, D.S. CLSM-FL: Clustering-based semantic federated learning in non-IID IoT environment. *IEEE Internet of Things Journal (IF: 9.9)*, 2024.
11. **Hunmin Lee**, J.B, H.L. Towards Utilization of Explainable Hand Motion Recognition in 3D Capacitive Proximity Sensing. *IEEE Sensors (IF: 4.7)*, 2024.

Employment

Fasikl Incorporated | Research and Development Intern 2025.05 - Current
- Designing an AI model for neural and EMG-based gesture decoding, and integration to a neural-computing interface.

University of Minnesota | Teaching and Research Assistant 2022.08 - 2025.05
- Research assistant: Developing a generalizable, personalizable AI model for a domain-specific environment

Outlier AI. | Contract Worker 2024.06 - 2024.08
- Human-in-the-loop supervision of Large Language Model

Georgia State University | Teaching and Research Assistant 2020.08 - 2022.05
- Research assistant: Building distributed AI architecture: 2 semesters

Korea Electric Power Corporation | Software Engineer Intern 2020.04 - 2020.08
- Advanced Metering Infrastructure (AMI) software maintenance team

Education

Computer Science, University of Minnesota — Minneapolis, MN, USA
Doctor of Philosophy (GPA: 3.97/4.0) — 2022 - 2026 *(expected)*

- Research area: Human-centered AI, AI in Distributed Environment

Computer Science, Georgia State University — Atlanta, GA, USA
Master of Science (GPA: 4.11/4.3) — 2020 - 2022

Computer Engineering, Chonnam National University — Gwangju, South Korea
Bachelor of Engineering (GPA: 3.90/4.5) — 2014 - 2020

Projects

Distributed Optimization via Federated Learning for Generalization and Personalization
*Supported by **Google DeepMind*** — 2025.08 - current

AI-powered HCI System for Rehabilitation
*Supported by **Fasikl Incorporated*** — 2024.08 - current

EAGER: Interpretable and Generalizable AI for Smart Manufacturing
Supported by National Science Foundation — 2022.08 - 2024.08

Multi-user interface for IoT using non-contact capacitive sensing
Supported by National Research Foundation of Korea — 2018.09 - 2020.04

Awards and Honors

- **Doctoral Dissertation Fellowship**, Awarded $26,000 as a Doctoral Dissertation Fellow, recognizing the university's most accomplished Ph.D. candidates. 2026
- **IOP Outstanding Reviewer Award**, Recognized for high-quality peer review contributions to the Journal of Neural Engineering. 2025
- **2^{nd} Prize**, AI/ICT device competition (Korean Ministry of Employment and Labor) 2024
- **3^{rd} Prize**, Social Data Analysis Competition (Dashin Group, Korea ESG Lab) 2023
- **1^{st} Prize**, AI-Hemp hackathon (Korean National Information Society Agency) 2022
- **1^{st} Prize**, 3-minute thesis award (Georgia State University), 2020
- **1^{st} Prize**, Asia open data challenge (GIGABYTE, Korea, Japan, Taiwan, Thailand) 2019
- **2^{nd} Prize**, AI-energy prediction competition (Korean Electric Power Corporation) 2019
- **2^{nd} Prize**, University Student Idea Contest (Korean Ministry of Small and Medium-sized Enterprises and Startups, Hyundai Motors) 2018

Skills

Languages: Korean (Native), English (Native-level).

Programming: Python, C, C++, MATLAB, Java, SQL, Go, Bash

ML/AI Frameworks: TensorFlow, PyTorch, Keras, Scikit-Learn, Deep Learning Toolbox

Tools: Git, Docker, Kubernetes, GCP

Services

Peer Reviewer (30+ papers): *IEEE Internet of Things Journal*, *IEEE Transactions on AI*, *IEEE Transactions on Mobile Computing*, *IEEE Transactions on Computational Social Systems*, *Journal of Neural Engineering*, etc.

Volunteer (2021-2022, Hanyang University): Mentored students in the Computer Science program.

Volunteer (2018-2019, Sekwang Special Education School): Taught English and Math and provided mentorship to visually impaired middle school students.